

HDOC DAY-42

Title: String Manipulation: Vowel Elimination, Repeating Character Search, and Anagram Checking

1. Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Remove all vowels from a string.
2. Find the first repeating lowercase alphabet in a string.
3. Check if two strings are anagrams of each other.
4. Exit.

2. Theoretical Background & Logic

1. Vowel Removal:

Iterates character-by-character through the input string. If `tolower(input[i])` is not equal to 'a', 'e', 'i', 'o', or 'u', the character is copied into the output array.

2. First Repeating Lowercase Alphabet:

Uses an array `freq[26]` initialized to 0 to keep track of counts of lowercase characters ('a' to 'z'). As soon as an element's frequency increments to 2, that character is

immediately returned as the first repeating lowercase alphabet.

3. Anagram Validation:

Two strings are anagrams if they contain the exact same characters with identical frequencies, regardless of order or case. A single frequency table tracks character increments for string 1 and decrements for string 2. If all frequency counts equal zero, the strings are anagrams.

3. Test Case Table

Test Case	Select ed Option	Input String(s)	Expected Output	Actual Output	Status
1	Remove Vowels	"Hello World"	"Hll Wrld"	"Hll Wrld"	Pass
2	First Repeating Lowercase	"programming"	First repeating character: 'r'	'r'	Pass

Test Case	Selected Option	Input String(s)	Expected Output	Actual Output	Status
3	Anagram Check	"listen" and "silent"	Are anagrams	Are anagrams	Pass
4	Anagram Check	"hello" and "world"	Are NOT anagrams	Are NOT anagrams	Pass
5	Exit	Choice 5	Program terminates cleanly	Program terminates cleanly	Pass

4. Source Code Listing

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#include <stdbool.h>

void read_string(const char *prompt, char str[], int max_size) {
    printf("%s", prompt);
    scanf(" %[^\\n]", str);
}
```

```

void remove_vowels(const char input[], char output[]) {
    int i = 0, j = 0;
    while (input[i] != '\0') {
        char ch = (char)tolower((unsigned char)input[i]);
        if (ch != 'a' && ch != 'e' && ch != 'i' && ch != 'o' && ch != 'u') {
            output[j++] = input[i];
        }
        i++;
    }
    output[j] = '\0';
}

```

```

char find_first_repeating_lowercase(const char str[]) {
    int freq[26] = {0};
    int i = 0;

    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z') {
            int index = str[i] - 'a';
            freq[index]++;
            if (freq[index] == 2) {
                return str[i];
            }
        }
        i++;
    }
    return '\0';
}

```

```

bool are_anagrams(const char str1[], const char str2[]) {
    int freq[256] = {0};
    int len1 = 0, len2 = 0;

    for (int i = 0; str1[i] != '\0'; i++) {
        if (!isspace((unsigned char)str1[i])) {
            char ch = (char)tolower((unsigned char)str1[i]);
            freq[(unsigned char)ch]++;
            len1++;
        }
    }
}

```

```

    }

    for (int i = 0; str2[i] != '\0'; i++) {
        if (!isspace((unsigned char)str2[i])) {
            char ch = (char)tolower((unsigned char)str2[i]);
            freq[(unsigned char)ch]--;
            len2++;
        }
    }

    if (len1 != len2) return false;

    for (int i = 0; i < 256; i++) {
        if (freq[i] != 0) return false;
    }

    return true;
}

int main(void) {
    char str1[200], str2[200], result[200];
    int choice;

    read_string("Enter primary string: ", str1, 200);

    do {
        printf("\n1. Remove Vowels\n2. First Repeating Lowercase\n3.
Anagram Check\n4. New String\n5. Exit\nChoice: ");
        if (scanf("%d", &choice) != 1) break;

        switch (choice) {
            case 1:
                remove_vowels(str1, result);
                printf("Result: %s\n", result);
                break;

            case 2: {
                char ch = find_first_repeating_lowercase(str1);
                if (ch != '\0') printf("First repeating lowercase: '%c'\n", ch);
                else printf("No repeating lowercase alphabet found.\n");
            }
        }
    } while (choice != 5);
}

```

```

        break;
    }

    case 3:
        read_string("Enter string 2: ", str2, 200);
        if (are_anagrams(str1, str2)) printf("ARE anagrams.\n");
        else printf("NOT anagrams.\n");
        break;

    case 4:
        read_string("Enter primary string: ", str1, 200);
        break;

    case 5: printf("Exiting...\n"); break;
    default: printf("Invalid choice!\n"); break;
}
} while (choice != 5);

return 0;
}

```

4. Execution Screenshots

```

kali@kali:~/HDOOC_Day42
$ mkdir HDOOC_Day42
$ cd HDOOC_Day42
$ nano day42.c
$ gcc day42.c -o day42
$ ./day42
Enter a primary string: programming

===== HDOOC DAY-42 STRING MENU =====
1. Remove all vowels from the string
2. Find the first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Enter a new primary string
5. Exit
Enter your choice (1-5): 1

Original String : programming
Without Vowels  : prgrmmg

===== HDOOC DAY-42 STRING MENU =====
1. Remove all vowels from the string
2. Find the first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Enter a new primary string
5. Exit
Enter your choice (1-5): 2

First repeating lowercase alphabet in 'programming': 'r'

```